How-To Library

Informatica

# Connecting to Oracle Database Cloud Service from Informatica Developer or PowerCenter

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# Abstract

You can configure an Oracle connection to connect to Oracle Database Cloud Service from Informatica® Developer or PowerCenter®. This article describes how to configure connectivity to Oracle Database Cloud Service.

# Supported Versions

- Informatica Developer 10.2 HotFix 1
- PowerCenter 10.2 HotFix 1

# Table of Contents

Overview. . . . . 2
Configure the ODBC Data Source on Windows. . . . . 3
Configuring the Oracle Client for Oracle Database Cloud Service to Read or Write Data. . . . . 6
Configuring SQLNET.ORA and TNSNAMES.ORA Files with Wallet Information. . . . . 7
Verifying Connectivity from Oracle Client to Oracle Database Cloud Service with SQL *Plus. . . . . 7
Creating an Oracle Connection from the Developer Tool. . . . . 7
Creating an Oracle Connection from PowerCenter. . . . . 8

# Overview

Oracle Database Cloud Service is an Oracle database that runs on a dynamic, on-demand cloud infrastruture. You can connect to Oracle Database Cloud Service from the Developer tool or PowerCenter Client to read and write data.

To import the metadata from Oracle Database Cloud Service and preview the data, you must create an ODBC data source for Oracle Database Cloud Service from the Developer tool or PowerCenter Client. Use the DataDirect ODBC Oracle Wire Protocol driver to create the data source.

To create an Oracle Database Cloud Service mapping from the Developer tool or PowerCenter Client, you must configure the Oracle Client with the Oracle Database Cloud Service database details. You can then use an Oracle connection in a mapping with the configured connection string to connect to Oracle Database Cloud Service from the Developer tool or PowerCenter Client.

This article describes the following tasks that you must perform to configure connectivity to Oracle Database Cloud Service from the Developer tool or PowerCenter Client:

1. Configure the ODBC data source for Oracle Database Cloud Service to import the metadata from Oracle Database Cloud Service.
2. To configure mappings using the Oracle connection to read data from or write data to Oracle Database Cloud Service, perform the following tasks:
    a. Configure the Oracle Client with the Oracle Database Cloud Service database details.
    b. Verify the connectivity from the Oracle Client to Oracle Database Cloud Service.
    c. Create an Oracle connection from the Developer tool or the PowerCenter Designer that you can use in a mapping to read data from or write data to Oracle Database Cloud Service.

# Configure the ODBC Data Source on Windows

You must create the ODBC data source for Oracle Database Cloud Service so that you can import the metadata and preview the data from the Developer tool or PowerCenter Client.

Create the data source on the machine where you installed the Developer tool or the PowerCenter Client. Configure the data source in the ODBC Data Source Administrator. You must create the 32-bit data source for the PowerCenter Client and the 64-bit data source for the Developer tool.

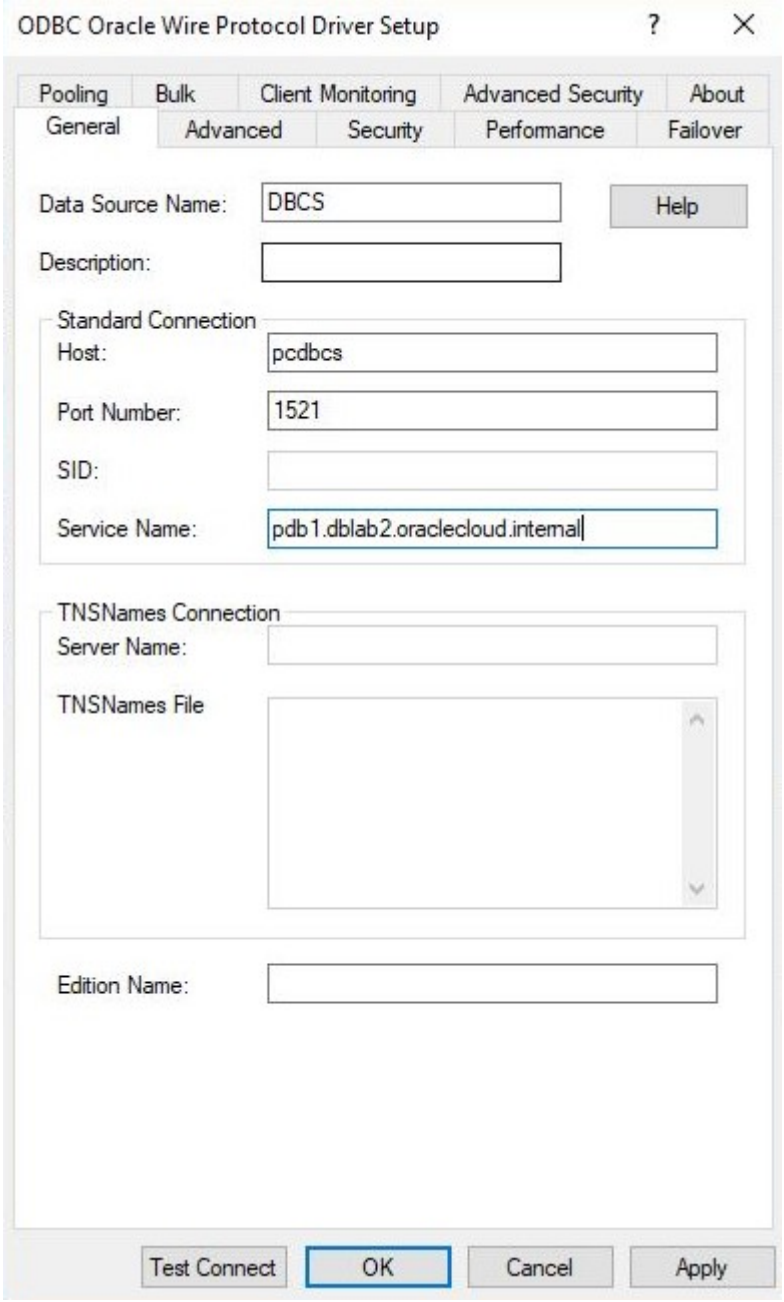
1. To start the ODBC Data Source Administrator wizard, perform one of the following tasks:
   - To create a 32-bit data source for the PowerCenter Client, open the Control Panel and click **Administrative Tools**. Then, double-click **ODBC Data Sources (32-bit)**.
     Alternatively, enter the following command at the command prompt:

     ```
     %windir%\SysWOW64\odbcad32.exe
     ```
   - To create a 64-bit data source for the Developer tool, open the Control Panel and click **Administrative Tools**. Then, double-click **ODBC Data Sources (64-bit)**.
     Alternatively, enter the following command at the command prompt:

     ```
     %windir%\System32\Odbcad32.exe
     ```

   The **ODBC Data Source Administrator** wizard appears.
2. On the **System DSN** tab, click **Add**.
3. In the **Create a New Data Source** dialog box, select **DataDirect 8.0 Oracle Wire Protocol** from the list of available drivers, and click **Finish**.
4. On the **General** tab of the **ODBC Oracle Wire Protocol Driver Setup** dialog box, enter the data source name, host name, port number, and service name for Oracle Database Cloud Service that you want to connect to.

The following image shows the `ODBC Oracle Wire Protocol Driver Setup` dialog box with the configured details:




5. On the **Security** tab, configure the following encryption properties:

**Encryption Method**

Get the value for SSL_VERSION in the database from your database administrator. For example, select *3-SSL3* if the SSL_VERSION in the database configuration is 3.0.

**Validate Server Certificate**

Enable this option if you want to validate the server certificate.

**Trust Store**

Enter the absolute path for the truststore `.p12` file.

**Trust Store Password**

Specify the password for the `.p12` truststore file.

**Key Store**

Enter the absolute path of the Oracle keystore `.p12` file.

**Key Store Password**

Specify the keystore password for `.p12` keystore file.

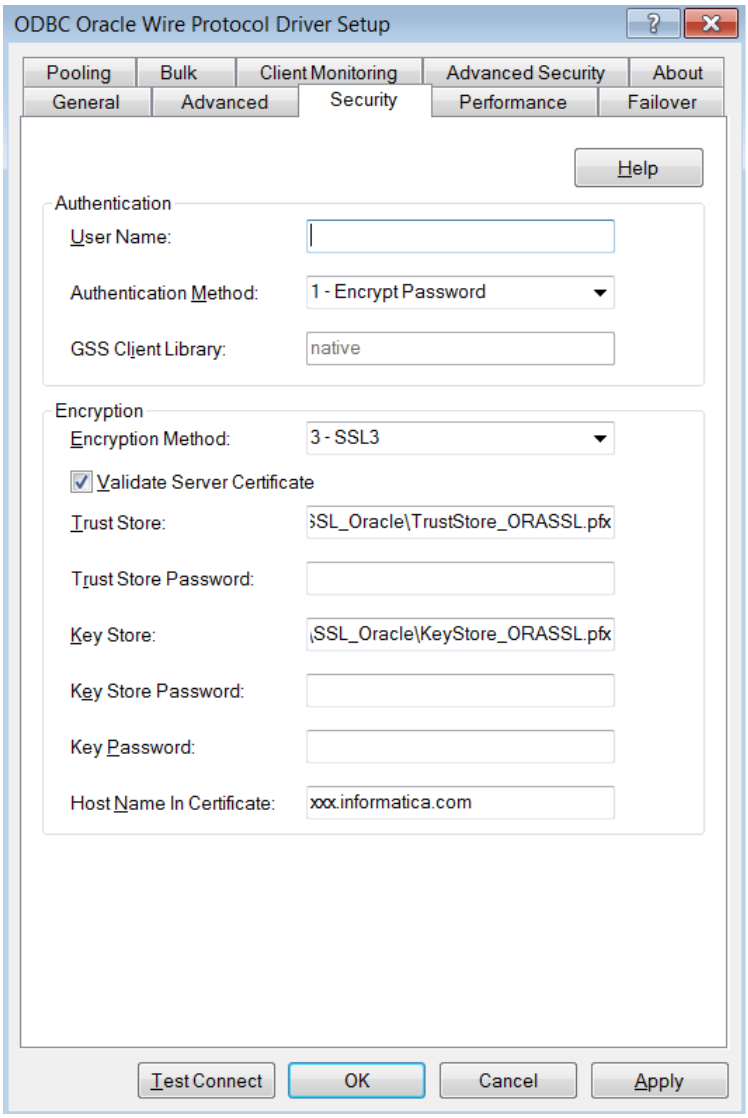
**Key Password**

Specify the key password. This field is optional.

**Host Name in Certificate**

You must get the host name details from the database administrator. Enter the name of the database server specified in the server SSL certificate.

The following image shows the DataDirect 8.0 Oracle Wire protocol setup where you configure the security settings:




6. Click **OK** to create the ODBC data source for Oracle Database Cloud Service.

## Configuring the Oracle Client for Oracle Database Cloud Service to Read or Write Data

After you create the data source, you must create an Oracle mapping from the Developer tool or PowerCenter Client to connect to Oracle Database Cloud Service to read or write data.

You must configure the Oracle Client to connect to Oracle Database Cloud Service. When you create a mapping, use an Oracle connection with the configured details to connect to Oracle Database Cloud Service.

## Configuring SQLNET.ORA and TNSNAMES.ORA Files with Wallet Information

Update the `sqlnet.ora` and `tnsnames.ora` files in the Oracle Client with the Oracle Database Cloud Service database details.

1. Navigate to the following Oracle Client directory:

   ```
   <Oracle Installation Directory>\Network\admin
   ```

2. Open the `sqlnet.ora` file, and edit to enter values similar to the following sample syntax: .

   ```
   WALLET_LOCATION = (SOURCE = (METHOD = file) (METHOD_DATA = (DIRECTORY="C:\Users\Documents
   \wallet")))
   ```

   ```
   SSL_SERVER_DN_MATCH=yes
   ```

   **Note:** Ensure that you set the wallet location to the directory that contains the `ewallet.p12` file.

3. Save and close the `sqlnet.ora` file.
4. Open the `tnsnames.ora` file and edit to enter the values similar to the following sample syntax:

   ```
   orassl = (DESCRIPTION = (ADDRESS = (PROTOCOL = TCPS)(HOST = <hostname>)(PORT =
   <port_number>)) (CONNECT_DATA = (SERVER = DEDICATED) (SERVICE_NAME = <service_name>))
   (SECURITY= (SSL_SERVER_CERT_DN="CN=ORASSL,OU=IT,O=infa,L=blr,ST=ka,C=IN")) )
   ```

5. Save and close the `tnsnames.ora` file.
6. Set the TNS_ADMIN environment variable to the directory where the `tnsnames.ora` and `sqlnet.ora` files reside.

## Verifying Connectivity from Oracle Client to Oracle Database Cloud Service with SQL *Plus

Verify that you can connect to Oracle Database Cloud Service from the Oracle Client machine.

1. Enter the following command from the Oracle Client machine:

   ```
   sqlplus <username>/<password>@<connection_string_for_Oracle>
   ```

   Use the connection string you defined in the `tnsnames.ora` file.

2. If you fail to connect to Oracle Database Cloud Service, verify that you have correctly entered the connectivity information.

   Contact your database administrator to resolve the connectivity issues.

## Creating an Oracle Connection from the Developer Tool

Use the Oracle connection to connect to Oracle Database Cloud Service from the Developer tool.

1. Open the Developer tool.
2. On the **Window** tab, select **Preferences**.
3. Select **Informatica** > **Connections**.
4. On the **Connections** window, expand **Databases**, and select **Oracle**.
5. Click **Add**.
6. In the Oracle connection properties, enter the Oracle Database Cloud Service database user name and password.
7. In the **Metadata Access Connection String** field, enter the following connection string details:

   ```
   jdbc:informatica:oracle://<hostname>:<portno>;
   ```

   ```
   ServiceName=<servicename>;EncryptionMethod=SSL;CryptoProtocolVersion=TLSv1.2,TLSv1.1,TLSv1;
   ```

```
KeyStore=<Absolute path to ewallet.p12>;KeyStorePassword=<keystorepassword>
```

8. In the **Data Access Connection String** field, add the TNS entry that you created for Oracle Database Cloud Service from the `tnsnames.ora` file.
9. Click **OK**.

You can use the configured Oracle connection to run a Oracle Database Cloud Service mapping from the Developer tool.

### *Creating an Oracle Connection from PowerCenter*

Create an Oracle connection to connect to Oracle Database Cloud Service from PowerCenter.

1. Log in to the Workflow Manager.
2. Click **Connections** > **Relational**

   The **Relational Connection Browser** dialog box appears.
3. Click **New**.

   The **Select Subtype** dialog box appears.
4. Select **Oracle** from the **Select Subtype** list.

   The **Relational Connection** dialog box appears.
5. Enter the connection properties, such as user name and password.
6. In the **Data Access Connection String** field, add the TNS entry that you defined in the `tnsnames.ora` file.

   For example, enter, `orassl`.
7. Click **OK**.

   The database connection appears in the **Connection Browser** list.

You can use the configured Oracle connection to run a Oracle Database Cloud Service session from PowerCenter.

## Authors

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